Posaconazole "Beats" Four CYP51 Binders by Docking but the Target Itself Is in Doubt: A Vina Benchmark of Mycobacterium tuberculosis CYP51 on 2CIB and the Repurposing Limits of Azole Anti-TB Activity

AI-assisted draft — NOT peer-reviewed, NOT published. All claims require independent laboratory verification. Do not cite as established evidence.

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Keywords: tuberculosis; CYP51; posaconazole; molecular docking; AutoDock Vina; azole; drug repurposing; Mycobacterium tuberculosis

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Abstract

Tuberculosis made about 10.8 million people ill in 2023 and killed about 1.25 million, and azole antifungals have shown real antimycobacterial activity — econazole and clotrimazole kill Mycobacterium tuberculosis H37Rv at 0.12 micrograms per millilitre — yet the target usually named for that activity, the sterol 14alpha-demethylase CYP51 (Rv0764c), sits in a bacterium that makes no sterols and lacks the rest of the sterol biosynthetic pathway, so whether CYP51 inhibition explains the killing is unproven. We ran the LeadForge pipeline live on 7 November 2026 (UniProtKB → ChEMBL → PubChem → RDKit → PAINS/ADMET → Vina docking): target M. tuberculosis CYP51 (UniProt P9WPP9, ChEMBL CHEMBL5090, 12 bioactivities), structure PDB 2CIB at 1.50 angstrom, grid centred exactly on the co-crystallised inhibitor CM6 ([–16.89, –2.19, 69.32], 6.7 angstrom from the haem centroid), and five ligands docked with AutoDock Vina 1.2.7 under an identical protocol. Posaconazole, the repurposed triazole comparator, scored –7.99 kcal/mol, beating every lead by 1.30–2.13 kcal/mol — fluconazole –6.44, CHEMBL521787 –6.69, CHEMBL507264 –6.64, CHEMBL491289 –5.86 — but the ranking among the leads is inverted against measured binding: fluconazole, the tightest binder by ChEMBL Kd (200 nM), scores third, behind two 5,000 nM binders, and all margins sit inside Vina's ±1–2 kcal/mol noise. The decisive limit is mechanistic, not numerical: azoles inhibit CYP51 by coordinating the haem iron, an interaction Vina cannot model, and posaconazole has no measured M. tuberculosis CYP51 affinity and no anti-TB MIC at all, so its docking win cannot be read as anti-TB activity. The falsifiable next step is a head-to-head CYP51 binding assay (spectral Kd) for posaconazole against fluconazole, with a pre-registered rule that a docking margin under 2 kcal/mol predicts no measurable affinity advantage; a measured posaconazole Kd tighter than fluconazole's 200 nM, or a CYP51-knockout strain that stays azole-sensitive, would refute the claim that the docking win is only a measurement of Vina's limits and of a target whose essentiality is unproven.

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1. Introduction

Tuberculosis (TB) remains one of the world's deadliest infectious diseases: according to the WHO Global Tuberculosis Report 2024, about 10.8 million people fell ill with TB in 2023 and about 1.25 million died. Standard treatment still takes months, and drug-resistant TB needs longer, more toxic regimens, so new drugs with new mechanisms are urgently needed — and repurposing an approved drug is the fastest route to one.

Azole antifungals are the oldest repurposing candidates in TB. Econazole and clotrimazole inhibit Mycobacterium smegmatis at MICs below 0.2 and 0.3 micromolar, better than rifampicin (1.2

micromolar) and isoniazid (36.5 micromolar) in the same assay, and both kill M. tuberculosis H37Rv at a MIC90 of 0.120 micrograms per millilitre with synergy against rifampicin and isoniazid. Econazole also kills nutrient-starved latent bacilli in vitro and reduces lung and spleen bacterial burden in latently infected mice, and in murine tuberculosis econazole alone reduced bacterial burden by 90% in lungs and spleen and was equipotent to rifampicin. Ketoconazole, the only systemically used azole tested in animals, reached an MIC of 8–16 micrograms per millilitre against H37Rv but showed no efficacy in the mouse model. The activity is real in a dish; the target is not settled.

The target usually named is CYP51 (Rv0764c), the M. tuberculosis sterol 14alpha-demethylase. It is a genuine enzyme: expressed in Escherichia coli, it 14alpha-demethylates lanosterol, 24,25-dihydrolanosterol and obtusifoliol, and its crystal structures with 4-phenylimidazole (PDB 1E9X, 2.1 angstrom) and fluconazole (PDB 1EA1, 2.21 angstrom) were the first structures of an authentic P450 drug target. Azoles bind it by coordinating the haem iron. But M. tuberculosis makes no sterols: it lacks squalene monooxygenase and oxidosqualene cyclase, the enzymes absolutely required for sterol biosynthesis, and the physiological function of its CYP51 remains obscure. The bacterium instead eats host cholesterol — cholesterol catabolism is required for persistence in animals — and the related P450 CYP121 binds azoles more tightly than CYP51 does, with Kd values in line with the antimycobacterial MICs. In the related actinomycete Streptomyces coelicolor, CYP51 is not essential for viability in vitro. So the azole killing of mycobacteria may run through CYP121, through several P450s at once, or through CYP51 — and docking against CYP51 alone cannot tell which.

Posaconazole is the strongest azole to test as a comparator: a broad-spectrum triazole with MIC90 of 0.5–1 microgram per millilitre against Candida and Cryptococcus and activity against fluconazole-resistant isolates, co-crystallised with the CYP51 of Trypanosoma cruzi and Trypanosoma brucei, and curative in murine Chagas disease. It has never been tested against M. tuberculosis CYP51 or against M. tuberculosis itself. The falsifiable hypothesis of this paper is that posaconazole docks to M. tuberculosis CYP51 more favourably than known CYP51 binders, and that the docking margin predicts a measurable binding advantage — a claim the paper then tests against the measured Kd values already in ChEMBL.

2. Methods

2.1 Literature basis

Sources were searched on 7 November 2026: the daily literature digest at `~/workspace/research-team/digests/2026-11-07.md`, PubMed and web search, ChEMBL (EBI), UniProtKB and RCSB PDB. Key sources: the WHO Global Tuberculosis Report 2024 for the TB burden figures; Bellamine et al. (1999) for the catalytic characterisation of M. tuberculosis CYP51; Podust et al. (2001) for the 1E9X/1EA1 crystal structures; McLean et al. (2002) for azole MICs against M. smegmatis and azole binding to CYP51 and CYP121; Ahmad et al. (2005, 2006) for econazole/clotrimazole activity against replicating and latent M. tuberculosis H37Rv; Ouellet et al. (2011) and Lamb et al. (2002) for the missing sterol pathway and the non-essentiality of CYP51 in S. coelicolor; Chen et al. (2010) for posaconazole bound to trypanosomatid CYP51; and the 2025 thiophene-thiazole proceeding for the only published docking score on this target. The posaconazole repurposing limits — no anti-TB MIC, no measured M. tuberculosis CYP51 affinity, hepatotoxicity and teratogenicity of long-term azole use, and MmpS5-MmpL5 efflux as the main azole-resistance mechanism in M. tuberculosis — are taken from Carroll & Parish (2015) and the azole literature cited there.

2.2 Molecular docking

The LeadForge pipeline was run live on 7 November 2026 (UniProtKB → ChEMBL → PubChem → RDKit → PAINS/ADMET → Vina). Target: M. tuberculosis CYP51, UniProt P9WPP9 (gene cyp51, Rv0764c, 451 amino acids, strain ATCC 25618 / H37Rv); ChEMBL CHEMBL5090, 12 bioactivities

retrieved. Thirteen compounds were ranked by composite score (similarity × 40 + Lipinski × 20 + Veber × 10 + PAINS × 10 + tox × 20). The top five — the query posaconazole plus four leads — were taken forward to docking. All four leads are known CYP51 binders with published ChEMBL Kd values; two have no preferred name in ChEMBL.

Docking was performed with AutoDock Vina 1.2.7, using the LeadForge protocol: the same structure, grid centre, box (22.5 angstrom cube), exhaustiveness (8) and pose count (3) for every ligand; best of three poses reported. The structure was M. tuberculosis CYP51, PDB 2CIB (1.50 angstrom, "High throughput screening and X-ray crystallography assisted evaluation of small molecule scaffolds for CYP51 inhibitors", co-crystallised ligand CM6: (2S)-2-[(2,1,3-benzothiadiazol-4-ylsulfonyl)amino]-2-phenyl-N-pyridin-4-ylacetamide), with the grid centred exactly on the centroid of the co-crystallised ligand CM6, chain A: [–16.89, –2.19, 69.32]. The centre was verified against the PDB file: the haem (HEM) centroid is [–17.48, –7.22, 64.88], 6.7 angstrom away, so the centre sits on the inhibitor above the haem, not on the haem cofactor — the grid-centre fault that has corrupted earlier runs (centring on a cofactor or glycan) did not occur here. Receptor preparation used RDKit (Gasteiger charges, simplified AutoDock atom types); ligand preparation used meeko. Vina's scoring noise is approximately ±1–2 kcal/mol; differences smaller than this cannot be treated as significant on scores alone.

One pipeline fault was caught and repaired before reporting. The first docking attempt for posaconazole failed: the Vina run hit the protocol's 900-second timeout (posaconazole is 700.79 Da with 12 rotatable bonds, and three other docking runs were competing for CPU on the same machine). The reference drug was re-docked alone at the identical centre, box and exhaustiveness, and succeeded in about 14 minutes of wall time with poses [–7.99, –7.80, –7.76] kcal/mol. The re-dock is reported as the posaconazole result; the timeout is documented here because a silent retry would hide a real property of the protocol — large, flexible ligands can exceed the per-ligand time budget.

2.3 PAINS screen

Every test compound was screened with the RDKit PAINS filter. Posaconazole carries one PAINS alert, `anil_di_alk_A(478)` — the N,N-dialkyl aniline substructure of its piperazine-linked phenyl ring. All four docked leads are PAINS-clean. A ToxGuard Tox21 kNN read-across panel (12 assays) was run for each compound: posaconazole's measured panel flags SR-ARE at 1.0; fluconazole flags nothing; the three unnamed leads flag only low read-across probabilities (highest NR-ER 0.276 for CHEMBL507264). These are similarity-based predictions, not measurements — and posaconazole is an approved drug, so its PAINS alert is a filter artefact for this scaffold class, not a liability finding.

2.4 Honesty limits

What this paper cannot show: docking is not activity; Vina noise is ±1–2 kcal/mol; no experimental data are presented; no molecular-dynamics refinement, binding-free-energy calculation, or re-docking validation of CM6 was performed; a single crystal structure (2CIB, one CYP51 conformation, holding a non-azole inhibitor) was used; and, most importantly, Vina cannot model the coordination bond between an azole nitrogen and the haem iron, which is the interaction that makes azoles CYP51 inhibitors — so every score reported here is missing the dominant binding term. Posaconazole has no measured affinity for M. tuberculosis CYP51 and no measured anti-TB MIC; the paper benchmarks a docking score, not a drug. The leads are already-known CYP51 binders, so no novelty of chemistry is claimed.

3. Results

3.1 Docking: the comparison table

Compound	ChEMBL Kd, Mtb CYP51	Best affinity (kcal/mol)	Poses (kcal/mol)	MW	logP	Lipinski violations	PAINS	Top tox read-across
Posaconazole (reference)	not measured	–**7.99**	–7.99, –7.80, –7.76	700.79	4.57	1	anil_di_alk_A(478)	SR-ARE 1.0 (measured)
CHEMBL521787	Kd 5,000 nM	–6.69	–6.69, –6.26, –6.21	277.41	3.26	0	clean	NR-ER 0.157
CHEMBL507264	Kd 5,000 nM	–6.64	–6.64, –6.41, –6.27	240.31	3.21	0	clean	NR-ER 0.276
Fluconazole (CHEMBL106)	Kd 200 nM	–6.44	–6.44, –6.29, –6.11	306.28	0.74	0	clean	none
CHEMBL491289	Kd 5,000 nM	–5.86	–5.86, –5.74, –5.61	231.29	2.77	0	clean	NR-AhR 0.157

All docking on PDB 2CIB (M. tuberculosis CYP51, 1.50 angstrom), grid centre [–16.89, –2.19, 69.32] (CM6 chain-A centroid), box 22.5 angstrom, exhaustiveness 8, Vina 1.2.7, 7 November 2026. Posaconazole: ChEMBL CHEMBL1397, PubChem CID 468595. Kd values are the ChEMBL bioactivities for target CHEMBL5090.

The headline, stated honestly: posaconazole "beats" every lead by 1.30–2.13 kcal/mol on docking — CHEMBL521787 by 1.30, CHEMBL507264 by 1.35, fluconazole by 1.55 and CHEMBL491289 by 2.13 kcal/mol — and it is the only margin in the table that clears Vina's ±1–2 kcal/mol noise, against the weakest lead. But three things cut against reading it as a discovery. First, posaconazole is by far the largest ligand in the set (700.79 Da, 12 rotatable bonds, against 231–306 Da for the leads), and Vina scores scale with the number of contacts a ligand makes, so a size advantage is expected and is not evidence of better binding. Second, the ranking among the four leads is inverted against measured binding: fluconazole, the tightest binder by Kd (200 nM), scores third, behind two 5,000 nM binders, and the 25-fold potency gap between fluconazole and the 5,000 nM binders corresponds to a docking gap of only 0.20–0.25 kcal/mol in the wrong direction. Third, the scored binding mode is the wrong one: azole inhibition of CYP51 is haem-iron coordination, which Vina cannot model at all, so the reference drug's score is missing the very interaction that defines its mechanism. Pose consistency does not rescue the ranking: posaconazole's three poses span only 0.23 kcal/mol and fluconazole's only 0.33, so both scores are stable — stably wrong about the order that matters.

No duplicate-entry fault occurred in this run: posaconazole (CHEMBL1397) does not appear among the 12 ChEMBL leads, unlike the macozinone duplicate caught in the DprE1 benchmark. One near-miss is worth recording: CHEMBL489925 (rank 7, Kd 5,000 nM, not docked — outside the top-5 cut) has the formula C19H15N5O3S2, the same as the co-crystallised ligand CM6, and its canonical SMILES is the CM6 scaffold itself — the crystal ligand's own ChEMBL entry sits in the lead list, one rank below the docking cut.

3.2 Comparison with published docking

One published docking score exists on this target: the thiophene-thiazole compound CTHTBN scored –8.8 kcal/mol against M. tuberculosis CYP51 (PDB 1E9X) with AutoDockTools 1.5.6. That number is not directly comparable to ours — different structure (1E9X, the 4-phenylimidazole complex, not 2CIB), different software version, and a different grid — but it sets the scale: published single-compound scores on this target run more negative than our best lead (–6.69) and somewhat more negative than posaconazole (–7.99), which is consistent with both numbers being dominated by ligand size and protocol rather than by measured affinity. No published Vina score for posaconazole

or fluconazole on 2CIB was found to benchmark against — which is itself a gap this paper fills: Vina 1.2.7 reference values for posaconazole and four known CYP51 binders on 2CIB are now on record for others to reproduce.

3.3 The standardisation / quality problem

Two things that could silently corrupt this result did not, and one thing that no structure check can catch did. The grid centre was verified against the PDB file and sits 6.7 angstrom from the haem centroid, on the co-crystallised inhibitor CM6, so the cofactor-centring fault is absent here. The duplicate-entry fault did not occur: posaconazole is absent from the lead list. The deeper problem is structural, not procedural: the reference drug is an azole whose mechanism is haem-iron coordination, and Vina has no term for metal coordination, so the score that "wins" by 1.30–2.13 kcal/mol is a score of van der Waals contacts and hydrogen bonds only. No grid-centre fix can repair that; only a method with a metal-coordination term (or a measured Kd) can.

4. Discussion

4.1 The reframe

The result reframes the paper from a discovery claim into a measurement of Vina's limits on a haem target — and into a question about the target itself. There is no lead here: the compound that "wins" is the repurposed comparator, it wins by a margin that its size alone can explain, the leads are ranked backwards against their measured Kd values, and the target's role in killing M. tuberculosis is unproven because the bacterium makes no sterols. The durable findings are the benchmark values for posaconazole on 2CIB (–7.99 kcal/mol, with full pose lists), the inversion of the lead ranking against Kd, the CM6-scaffold entry sitting in the lead list, and the demonstration that on a haem-coordination target, non-covalent docking ranks known binders in the opposite order to their measured affinity.

4.2 Why the "win" must not be over-read

Four reasons. First, the margins against the three best leads (1.30–1.55 kcal/mol) are inside or at the edge of Vina's ±1–2 kcal/mol noise; only the 2.13 kcal/mol margin against the weakest lead clears it. Second, the ranking among the leads is inverted against measured Kd — a 200 nM binder scores below two 5,000 nM binders. Third, the winner is 2.3–3.0 times the molecular weight of the leads, and Vina rewards contact count. Fourth, the scored binding mode is the wrong one: haem-iron coordination is absent from every score reported here. A docking "win" for an azole on a haem enzyme, on a structure holding a non-azole inhibitor, is not a discovery; it is a measurement of what the method leaves out.

4.3 The repurposing limits are the experiment's design constraint, not a reason to stop

Posaconazole is not an anti-TB drug and this paper does not make it one. There is no published anti-TB MIC for posaconazole, no measured affinity for M. tuberculosis CYP51, and no animal data in TB; the Chagas data (cure in 50–100% of acutely infected mice) is against a different parasite's CYP51, not M. tuberculosis. Long-term azole use carries hepatotoxicity and teratogenicity, and M. tuberculosis resists azoles chiefly by MmpS5-MmpL5 efflux — both reasons a repurposing claim would need whole-cell and animal data before it could be taken seriously. The PAINS alert on posaconazole (`anil_di_alk_A(478)`) is a filter artefact on an approved drug and does not block validation, but it is reported because the filter output is part of the record. The validation assay must still include a counter-screen against human CYP51 and CYP121, because selectivity — not CYP51 affinity alone — is what would make an azole a TB drug rather than a broad P450 poison.

4.4 Testable hypotheses

• Posaconazole binds recombinant M. tuberculosis CYP51 with a spectral Kd tighter than fluconazole's 200 nM — the docking predicts a measurable advantage; a Kd equal to or weaker than 200 nM would refute the noise interpretation of the 1.55 kcal/mol margin.

• A Vina re-dock of the co-crystallised ligand CM6 (CHEMBL489925) into 2CIB reproduces its crystallographic pose within 2 angstrom RMSD (protocol self-validation; not yet run).

• A CYP51-knockout (Rv0764c deletion) strain of M. tuberculosis H37Rv shows unchanged azole sensitivity — which would show CYP51 is not the target of azole killing, and would reframe the whole benchmark as a measurement on a bystander enzyme.

• Docking the same five ligands against M. tuberculosis CYP121 (PDB 2IJ7, fluconazole complex) ranks econazole/clotrimazole-class binders in agreement with their measured Kd values — testing whether the inversion is specific to CYP51 or general to haem targets under Vina.

4.5 Proposed validation (university-lab feasible, 8–12 weeks)

Stage 1 (weeks 1–2): spectral binding assay (Soret shift titration) for posaconazole, fluconazole and CHEMBL521787 against purified recombinant M. tuberculosis CYP51, same plate, n = 3, with a pre-registered 2-fold decision rule and the rule that a docking margin under 2 kcal/mol predicts no measurable affinity advantage. Stage 2 (weeks 3–6): whole-cell M. tuberculosis H37Rv MIC assay (resazurin microtiter assay) for posaconazole, with econazole as the positive azole control (MIC90 0.120 micrograms per millilitre) and fluconazole as the negative control, plus a human CYP51 counter-screen for selectivity. Stage 3 (weeks 7–12): if Stage 1 shows any genuine affinity advantage, a CYP51-knockout sensitivity test and a CYP121 docking comparison (hypothesis 4); if Stage 1 shows none, the project stops at the benchmark and the paper stands as a negative result. Total compound cost is the binding constraint; posaconazole and fluconazole are approved drugs, the unnamed leads are research chemicals, and none may be presented as a treatment.

5. Limitations

• Docking only; no experimental data. 2. Three of the four "winning" margins (1.30–1.55 kcal/mol) are inside Vina's ±1–2 kcal/mol noise and cannot be interpreted as real advantages. 3. Vina cannot model haem-iron coordination, so the azole mechanism is absent from every score reported here; the structure used (2CIB) holds a non-azole inhibitor. 4. No molecular-dynamics refinement, binding-free-energy calculation, metal-aware docking, or re-docking validation of CM6 was performed. 5. A single crystal structure (2CIB, one CYP51 conformation) was used; induced-fit effects are not modelled. 6. Posaconazole has no measured M. tuberculosis CYP51 affinity and no anti-TB MIC — the comparator is repurposed on the basis of antifungal and Chagas data only. 7. The target's essentiality is unproven: M. tuberculosis makes no sterols, lacks the sterol pathway, and the related CYP51 of S. coelicolor is non-essential in vitro. 8. The first posaconazole docking attempt timed out at the 900-second protocol limit under CPU contention and had to be re-run alone; the reported score comes from the re-dock at identical settings. 9. The PAINS alert and Tox21 read-across values are filter and similarity outputs, not measurements.

6. Conclusion

Posaconazole docks to M. tuberculosis CYP51 at −7.99 kcal/mol, 1.30–2.13 kcal/mol better than four known binders — but the leads rank backwards against their measured Kd values, the winner's size alone can explain its margin, Vina cannot see the haem coordination that defines azole binding, and the bacterium makes no sterols, so the target itself is in doubt. The honest reading is that on a haem target with an unproven role in TB, non-covalent docking measures the method, not the molecule — and the next experiment is a spectral Kd assay with a pre-registered 2 kcal/mol rule, plus a CYP51-knockout sensitivity test, not more Vina docking.

Acknowledgments

This manuscript was prepared with AI assistance (Lapis) for drafting, literature search and docking computations. The author reviewed the manuscript and takes responsibility for its content. No funding was received. The author thanks the ChEMBL, UniProtKB, RCSB PDB and PubChem teams for free, open data.

References

• World Health Organization. Global Tuberculosis Report 2024. Geneva: WHO; 2024. (Approximately 10.8 million people fell ill with TB in 2023 and about 1.25 million died.)

• Eberhardt J, Santos-Martins D, Tillack AF, Forli S. AutoDock Vina 1.2.0: new docking methods, expanded force field, and Python bindings. *J Chem Inf Model.* 2021;61(8):3891–3898. PMID: 34278794.

• Bellamine A, Mangla AT, Nes WD, Waterman MR. Characterization and catalytic properties of the sterol 14alpha-demethylase from Mycobacterium tuberculosis. *Proc Natl Acad Sci U S A.* 1999;96(16):8937–8942. DOI: 10.1073/pnas.96.16.8937.

• Podust LM, Poulos TL, Waterman MR. Crystal structure of cytochrome P450 14alpha-sterol demethylase (CYP51) from Mycobacterium tuberculosis in complex with azole inhibitors. *Proc Natl Acad Sci U S A.* 2001;98(6):3068–3073. PMID: 11248033. DOI: 10.1073/pnas.061562898.

• RCSB Protein Data Bank. M. tuberculosis CYP51 in complex with the inhibitor CM6 (high-throughput screening scaffold). PDB ID 2CIB, 1.50 angstrom. https://www.rcsb.org/structure/2CIB (accessed 7 November 2026).

• McLean KJ, Marshall KR, Richmond A, Hunter IS, Fowler K, Kieser T, Gurcha SS, Besra GS, Munro AW. Azole antifungals are potent inhibitors of cytochrome P450 mono-oxygenases and bacterial growth in mycobacteria and streptomycetes. *Microbiology.* 2002;148(10):2937–2949. DOI: 10.1099/00221287-148-10-2937.

• Ahmad Z, Sharma S, Khuller GK. In vitro and ex vivo antimycobacterial potential of azole drugs against Mycobacterium tuberculosis H37Rv. *FEMS Microbiol Lett.* 2005;251(1):19–22. DOI: 10.1016/j.femsle.2005.07.022.

• Ahmad Z, Sharma S, Khuller GK. The potential of azole antifungals against latent/persistent tuberculosis. *FEMS Microbiol Lett.* 2006;258(2):200–203. DOI: 10.1111/j.1574-6968.2006.00224.x.

• Ouellet H, Johnston JB, Ortiz de Montellano PR. Cholesterol catabolism as a therapeutic target in Mycobacterium tuberculosis. *Trends Microbiol.* 2011;19(11):530–539. PMID: 21924910. DOI: 10.1016/j.tim.2011.07.009.

• Lamb DC, Fowler K, Kieser T, Manning N, Podust LM, Waterman MR, Kelly DE, Kelly SL. Sterol 14alpha-demethylase activity in Streptomyces coelicolor A3(2) is associated with an unusual member of the CYP51 gene family. *Biochem J.* 2002;364(2):555–562. DOI: 10.1042/BJ20011380.

• Matsuura K, Yoshioka S, Tosha T, Hori H, Ishimori K, Kitagawa T, Morishima I, Kagawa N, Waterman MR. Structural diversities of active site in clinical azole-bound forms between sterol 14alpha-demethylases (CYP51s) from human and Mycobacterium tuberculosis. *J Biol Chem.* 2005;280(10):9088–9096. PMID: 15611056. DOI: 10.1074/jbc.m413042200.

• Chen CK, Leung SSF, Guilbert C, Jacobson MP, McKerrow JH, Podust LM. Structural characterization of CYP51 from Trypanosoma cruzi and Trypanosoma brucei bound to the antifungal drugs posaconazole and fluconazole. *PLoS Negl Trop Dis.* 2010;4(4):e651. DOI: 10.1371/journal.pntd.0000651.

• Pfaller MA, Messer SA, Boyken L, Rice C, Tendolkar S, Hollis RJ, Diekema DJ. In vitro activities of posaconazole, fluconazole, itraconazole, voriconazole, and amphotericin B against a large collection of clinically important molds and yeasts. *Antimicrob Agents Chemother.* 2006;50(6):2009–2015. PMID: 16723559. DOI: 10.1128/AAC.00163-06.

• Torres HA, Hachem RY, Chemaly RF, Kontoyiannis DP, Raad II. Posaconazole: a broad-spectrum triazole antifungal. *Lancet Infect Dis.* 2005;5(12):775–785. PMID: 16310149. DOI: 10.1016/S1473-3099(05)70297-8.

• Carroll P, Parish T. Deletion of cyp125 confers increased sensitivity to azoles in Mycobacterium tuberculosis. *PLoS One.* 2015;10(7):e0133129. DOI: 10.1371/journal.pone.0133129.

• McLean KJ, et al. Multiple drug binding modes in Mycobacterium tuberculosis CYP51B1. *J Inorg Biochem.* 2020;205:110994. DOI: 10.1016/j.jinorgbio.2020.110994.

• Synthesis, characterization, DFT study, and in silico evaluation of a thiophene-thiazole scaffold as a potential Mycobacterium tuberculosis CYP51 inhibitor. *Proceedings.* 2025;18(1):58. https://www.mdpi.com/2673-4583/18/1/58 (accessed 7 November 2026).

• ChEMBL database, European Bioinformatics Institute. Target: sterol 14alpha-demethylase, M. tuberculosis (CHEMBL5090); compounds posaconazole (CHEMBL1397), fluconazole (CHEMBL106), miconazole (CHEMBL91), clotrimazole (CHEMBL104), econazole (CHEMBL808); bioactivities accessed 7 November 2026. https://www.ebi.ac.uk/chembl/.

• Ahmad Z, Sharma S, Khuller GK. Azole antifungals as novel chemotherapeutic agents against murine tuberculosis. *FEMS Microbiol Lett.* 2006;261(2):181–186. DOI: 10.1111/j.1574-6968.2006.00350.x.

Data availability

All docking scores reported here are in `hidden_files/dock-HP-021-2026-11-07.json` (pipeline run) and `hidden_files/dock-HP-021-2026-11-07-posaconazole.json` (posaconazole re-dock). ChEMBL IDs: posaconazole CHEMBL1397 (PubChem CID 468595), fluconazole CHEMBL106, CHEMBL521787, CHEMBL507264, CHEMBL491289, CHEMBL489925 (CM6 scaffold). Target: UniProt P9WPP9, ChEMBL CHEMBL5090. PDB ID 2CIB. No additional data were generated.